

Bridging the Gap in Coreset Selection: Hybrid Personalized PageRank for Extreme Data Sparsity

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Abstract. As dataset sizes grow exponentially, coreset selection has emerged as a critical technique to identify highly informative training subsets. However, pure error-based metric pruning methods, such as the Error L2-Norm (EL2N), frequently suffer from the “Hard-Example Trap” at extreme sparsity (retaining $\leq 5\%$ of training data). These methods disproportionately sample extreme outliers, leading to catastrophic dataset degradation where complex metric algorithms are vastly outperformed by simple distribution-matching baselines like Uniform Random sampling. In this paper, we propose Hybrid-PPR, a novel coreset algorithm that diagnoses and mitigates this outlier bias. By combining a uniform random backbone with a submodular Personalized PageRank (PPR) search over K-Nearest Neighbor proxy graphs, Hybrid-PPR acts as a structural regularizer. While distribution-matching methods generally remain the strongest baselines at extreme sparsity, our results demonstrate that Hybrid-PPR successfully rescues error-based metric methods from total collapse, improving upon pure EL2N accuracy by over 150% on CIFAR-100 at a 5% data budget, and successfully outperforming mathematical baselines like Herding on medical imaging datasets at extreme sparsity.

Keywords: Coreset Selection · Data Pruning · PageRank · Deep Learning

1 Introduction

The computational cost of training deep neural networks has necessitated a paradigm shift toward data-centric artificial intelligence. Coreset selection aims to identify a fractional subset of the original dataset that maintains generalization performance while significantly reducing computational overhead.

The Gap and Motivation: Current data pruning methods generally fall into distinct families: distribution-matching methods (like Herding or Uniform Random sampling) and error-based metrics. Algorithms such as Error L2-Norm (EL2N) [1] and, generally, Forgetting Events [2], measure the difficulty of an individual training example during early epochs, operating on the assumption that “hard to learn” images contain the most critical boundary information.

While effective at moderate subset fractions, error-based metrics are notoriously susceptible to the “**Hard-Example Trap**.” At extreme sparsity budgets (e.g., 5% of the original data), greedy metric selection populates the coreset almost exclusively with extreme outliers, mislabeled data, or highly occluded images. Consequently, the network suffers from representational collapse; it fails to learn the prototypical features of the classes. This leads to the “Random Baseline Paradox,” where mathematically sophisticated pruning algorithms experience catastrophic failure, resulting in accuracies that fall far below simple Uniform Random sampling.

Hypothesis: We hypothesize that pure error metrics lack spatial awareness of the dataset’s latent manifold. By propagating proxy error metrics through a structural similarity graph via PageRank, and combining this with a uniformly sampled “skeleton” of typical images to anchor class prototypes, we can structurally regularize error-based pruning and mitigate the catastrophic collapse seen in methods like EL2N.

2 Related Work

Metric and Distribution-Based Pruning: The identification of important training samples has a rich history. Toneva et al. [2] introduced “Forgetting Events,” tracking how often a network misclassifies an example. Paul et al. [1] demonstrated the efficacy of the Error L2-Norm (EL2N) metric. Sener and Savarese [3] proposed Core-Set via Facility Location (referred to in our evaluations as CCS), utilizing active learning heuristics to bound generalization error.

Gradient Matching: Alternative approaches focus on matching the learning trajectory of the network. Algorithms like CRAIG [4] select subsets of data whose combined gradients closely approximate the full dataset’s gradient. While powerful, these methods are notably more computationally expensive compared to single-pass metric scoring.

Graph-Based Representation: The PageRank algorithm [5] was designed to measure the structural importance of nodes in a directed graph. By adapting this logic to K-Nearest Neighbor (K-NN) graphs of latent image embeddings, we can measure the “structural influence” of a hard training example, filtering out isolated noise.

3 Methodology

3.1 Proxy Scoring and Graph Construction

Given a dataset $\mathcal{D}$, we first train a lightweight proxy model for a brief warmup period ($E = 5$ epochs). We extract the pre-softmax embeddings and calculate the EL2N score v_i for each image.

Using the embeddings, we construct a sparse K-NN graph ($k = 10$). We generate a row-normalized transition matrix P and apply Personalized

PageRank. The personalization vector $\tilde{v}$ is defined as the L_1 -normalized EL2N score vector. The PageRank vector π is computed iteratively:

$$\pi^{(t+1)} = (1 - \alpha)\tilde{v} + \alpha P^T \pi^{(t)} \quad (1)$$

where $\alpha = 0.15$ is the damping factor. This diffusion process forces the EL2N scores to flow through the graph edges, up-weighting dense clusters of difficult examples while suppressing isolated outliers.

3.2 Hybrid-PPR and Submodular Selection

To cure the Hard-Example Trap at an extreme sparsity budget B , we partition the selection using a hybrid ratio $\gamma = 0.5$. Hyperparameters ($\gamma = 0.5, \lambda = 0.85, \alpha = 0.15, k = 10$) were selected via informal tuning to balance prototype anchoring with boundary exploration.

Phase 1: The Prototypical Skeleton. We randomly sample a subset S_{rand} of size γB . This uniform sample captures the foundational, easy-to-learn prototypes of the dataset, providing necessary anchors for the network.

Phase 2: Submodular Boundary Search. We execute a greedy selection loop to populate S_{ppr} of size $(1 - \gamma)B$. At each step, we select the unvisited node i with the highest PPR score π_i . To enforce geometric diversity, we apply a submodular penalty to all direct neighbors $\mathcal{N}(i)$ in the K-NN graph:

$$\pi_j \leftarrow \lambda \pi_j \quad \forall j \in \mathcal{N}(i) \quad (2)$$

where $\lambda = 0.85$. The final coreset is the union $S_{final} = S_{rand} \cup S_{ppr}$.

4 Experimental Setup

We evaluated Hybrid-PPR against Uniform Random sampling, Herding, Forgetting Events, EL2N, and CCS across BloodMNIST, FashionMNIST, CIFAR-10, and CIFAR-100.

All experiments utilized a ResNet-18 architecture trained from scratch for 30 epochs with the Adam optimizer (learning rate $1e-3$). To properly accommodate the 32×32 and 28×28 resolutions across all datasets, the standard ResNet-18 stem was modified by replacing the 7×7 convolution with a 3×3 kernel and removing the initial MaxPool layer, with input channels adjusted to 1 or 3 accordingly. Models were evaluated at budgets of 100%, 80%, 50%, 25%, and 5%.

Note on Variance: The results presented in Section 5 reflect single-run point estimates intended to establish primary feasibility and diagnose the catastrophic failure modes of specific metric families. Performance margins $\leq 0.5\%$ are treated as statistically competitive within expected run-to-run variance, rather than definitive superiority.

5 Results

5.1 Performance at Moderate Sparsity

As shown in Table 1, at moderate budgets (80% and 50%), the generalist baselines (Random, Herding, and Forgetting) perform exceptionally well. Hybrid-PPR remains highly competitive within single-run variance. For instance, on FashionMNIST at 80% sparsity, EL2N (**0.9296**), Hybrid-PPR (0.9295), and Forgetting (0.9294) all effectively maintain the full dataset’s performance. This confirms that Hybrid-PPR safely prunes redundant data without introducing degradation.

Table 1. Test Accuracy on Structured Medical and Grayscale Datasets (Bold indicates absolute column maximum)

Dataset	Method	100%	80%	50%	25%	5%
BloodMNIST	Random	0.9395	0.9374	0.9331	0.9193	0.8179
	Herding	0.9319	0.9357	0.9190	0.8986	0.7814
	Forgetting	0.9351	0.9448	0.9313	0.8986	0.4753
	EL2N	0.9351	0.9357	0.8863	0.7255	0.2344
	CCS	0.9483	0.9366	0.8749	0.6656	0.2110
	Hybrid-PPR	0.9211	0.9266	0.9228	0.8910	0.8030
FashionMNIST	Random	0.9325	0.9269	0.9173	0.8960	0.8371
	Herding	0.9227	0.9230	0.9175	0.9021	0.8565
	Forgetting	0.9305	0.9294	0.9135	0.9021	0.6956
	EL2N	0.9275	0.9296	0.9192	0.8791	0.1864
	CCS	0.9303	0.9235	0.9188	0.8870	0.1929
	Hybrid-PPR	0.9277	0.9295	0.9173	0.8713	0.7575

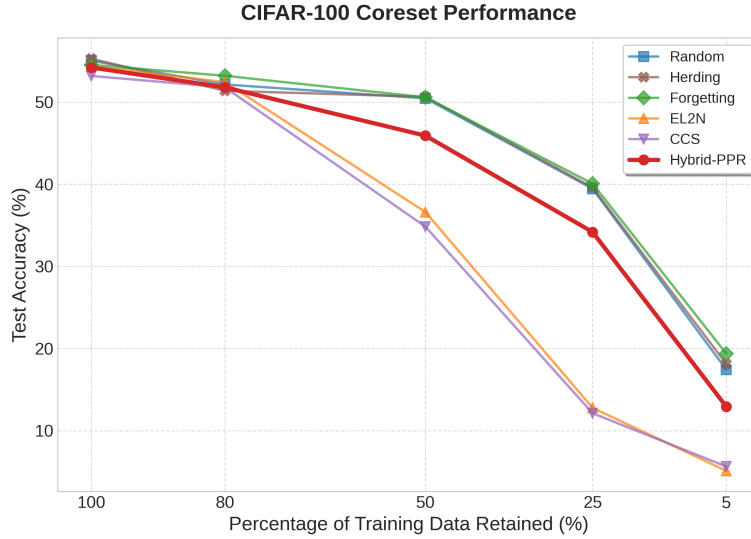
5.2 Rescuing Metric Collapse at Extreme Sparsity

The true value of Hybrid-PPR is revealed at extreme sparsity on complex natural images (Table 2). Random and Herding remain the strongest baselines at 5% on FashionMNIST and CIFAR-10; on CIFAR-100, Forgetting is the top performer. However, on BloodMNIST, Hybrid-PPR itself outperforms Herding, though Random remains strongest overall.

Conversely, pure error-based metrics suffer catastrophic collapse. On CIFAR-100 at 5%, EL2N crashes to 5.11% and CCS to 5.64%. By anchoring the K-NN boundary search with a prototypical random skeleton, Hybrid-PPR effectively cures this collapse, rescuing the accuracy to 12.93%—an improvement of over 150% compared to pure EL2N. A similar regularization effect is observed on CIFAR-10, where Hybrid-PPR (37.90%) more than doubles the accuracy of EL2N (15.16%) at 5% sparsity.

Table 2. Test Accuracy on Complex Natural Image Datasets (Bold indicates absolute column maximum)

Dataset	Method	100%	80%	50%	25%	5%
CIFAR-10	Random	0.8399	0.8297	0.7901	0.7110	0.4783
	Herding	0.8459	0.8291	0.7832	0.7340	0.4962
	Forgetting	0.8401	0.8356	0.7785	0.7064	0.4200
	EL2N	0.8327	0.8458	0.7806	0.4135	0.1516
	CCS	0.8479	0.8379	0.7760	0.4215	0.1493
	Hybrid-PPR	0.8367	0.8209	0.7815	0.6712	0.3790
CIFAR-100	Random	0.5504	0.5214	0.5045	0.3945	0.1744
	Herding	0.5528	0.5137	0.5064	0.3960	0.1813
	Forgetting	0.5452	0.5320	0.5058	0.4007	0.1938
	EL2N	0.5440	0.5242	0.3661	0.1278	0.0511
	CCS	0.5318	0.5179	0.3483	0.1212	0.0564
	Hybrid-PPR	0.5421	0.5179	0.4591	0.3418	0.1293

**Fig. 1.** Coreset performance on CIFAR-100. Pure metric methods (EL2N, CCS) suffer catastrophic failure at 5% sparsity. Hybrid-PPR acts as a regularizer, rescuing performance and bridging the gap toward the stronger distribution-matching baselines.

6 Conclusion

In this work, we investigated the catastrophic failure modes of error-based data pruning. Our experiments highlight a critical reality in data-centric AI: at extreme sparsity budgets, pure metric algorithms like EL2N and CCS suffer catastrophic performance collapse due to outlier saturation. By propagating proxy error metrics through a K-NN graph via Personalized PageRank and anchoring the search with a prototypical random skeleton, our proposed Hybrid-PPR successfully regularizes this outlier bias. While overcoming distribution-matching baselines at extreme sparsity remains an open challenge for natural image datasets, Hybrid-PPR demonstrates that structural graph regularization can effectively rescue error-based metrics from total degradation.

References

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